

SAMPLE PROJECT DOCUMENTATION

ComicCraft - AI Comic Story Creator using Gemini Models • Team ID: SWTID-2026-6777

Skill Wall / SmartBridge Project-Based Learning Student Workbook

Project Identification Summary

ComicCraft is a Generative AI comic story creator for aspiring storytellers, students, and hobbyists. A user enters a short story idea, a main character, a setting, a mood, and an art direction, and ComicCraft turns that spark into a five-frame comic with a title, narration, dialogue, and art, powered by Google Gemini models and a FastAPI web application.

Project Metadata Record

Field	Details
Project Title	ComicCraft - AI Comic Story Creator using Gemini Models
Team Identifier (Team ID)	SWTID-2026-6777
Team Members	Pravinkumar P, Mohammed Jamil Fiyaz J, Vishal M, Thirumurugan A, Sakthibavan R
Submission Date	30 September 2026
Track / Domain	Generative AI, Web Systems & Applied Machine Learning
Backend & Runtime	Python, FastAPI Framework, Uvicorn ASGI Server
AI Engines	Google Gemini models (story, characters, dialogue, image prompts); optional image-generation service for live art
Frontend & Templating	HTML, CSS, JavaScript, Jinja2 Templates
Security Governance	Zero-hardcoded-secrets policy (API keys supplied through environment variables)

Project Primary Objectives

- **Story Idea to Comic:** Turn a short story spark into a complete five-frame comic with a title and a clear beginning, middle, and end.
- **Character and Setting Development:** Build the story around the main character and setting the user provides.
- **Narration and Dialogue:** Write frame-by-frame narration and character dialogue that fit the chosen mood.
- **Art Direction:** Produce scene art and image prompts that follow the selected art style (for example, Storybook).
- **Keep and Iterate:** Let users download their comic, make another, or try a fresh spark with no account required.

Student Declaration

This documentation reflects the technical specification, architectural design, interface contract, and implementation approach developed by Team SWTID-2026-6777 for the ComicCraft system.

Project Overview & Problem Statement

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1. Problem Being Addressed

Many people have story ideas but struggle to turn them into finished comics. Common challenges include:

- **Blank-Page Paralysis:** Starting a story is hard, and writer's block stops many ideas before the first panel.
- **Panel Planning:** Breaking a story into frames with narration and dialogue takes planning skill.
- **Art Barrier:** Drawing every panel takes time and artistic skill that many storytellers do not have.
- **Scattered Tools:** Writing, scripting, and illustration are usually done in separate tools with no single guided workflow.

2. Target Users & Stakeholders

User Group	Observed Need	ComicCraft Impact
Aspiring Comic Creators	Turn ideas into visual stories without drawing skills.	Story, dialogue, and art generated from one short spark.
Students	Creative writing and storytelling assignments.	A guided five-frame structure that shows how a story unfolds.
Teachers & Educators	Engaging visual material for lessons and creative exercises.	Quick comic drafts from a classroom prompt.
Hobbyists & Content Creators	Fast prototypes of characters and scenes.	Ready-to-use frames and image prompts to build on.

3. Why Users Need ComicCraft

General-purpose chatbots return plain text and leave layout, dialogue, and art to the user. ComicCraft is built around a comic-creation loop: **Dream** (enter a story spark), **Draw** (generate frames with narration, dialogue, and art), and **Keep** (download or make another).

4. Expected Benefits

- **Faster Starts:** A short spark becomes a complete story outline and script in one request.
- **Lower Barrier to Creativity:** Users get finished-looking frames without drawing skills.
- **Consistent Structure:** Every comic has the same five-frame shape, which makes results easy to read and compare.
- **Privacy and Simplicity:** No account is needed, and the story stays on the user's device.

5. Project Scope & Boundaries

ComicCraft is a standalone web application served by FastAPI. It accepts a story spark with character, setting, mood, and art direction, and returns a five-frame comic. External dependencies are limited to Google Gemini models for story generation and, optionally, an image-generation service for live art. Without API keys, the application runs in a local demo mode with generated preview text and sample art. User accounts, a persistent database, and deployment to a public host are outside the current MVP.

Project Scenarios & User Use Cases

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The following scenarios show how ComicCraft processes real user inputs. The main example is the story shown in the interface screenshots.

Scenario 1: Creating a Comic from a Story Spark

User Input: Story spark: "A curious fox finds a lantern that reveals a hidden path through the moonlit forest." | Main character: Fern | Setting: Moonlit forest | Mood: Adventurous | Art direction: Storybook.

System Processing: The web form submits the inputs to the FastAPI generation route, which validates them and asks the Gemini model for a five-frame story.

Output: A comic titled "Fern and the Moonlit Forest Mystery" with five frames, each with a title, art, narration, and dialogue (for example, frame 1 "The first glimmer", frame 2 "A clue in the wild", frame 3 "Into the unknown").

Scenario 2: Changing Mood and Art Direction

User Input: The same story spark with a different story mood or art direction chosen from the dropdown menus.

Output: A new comic whose tone, narration, and visual style follow the new selections.

Scenario 3: Trying a Fresh Spark

User Input: The user clicks "Try a fresh spark" on the form when unsure what to write.

Output: The form is filled with an example story that can be edited or submitted as it is.

Scenario 4: Downloading and Making Another

User Input: After reading the comic, the user clicks "Download your comic" or "Make another".

Output: The comic is saved to the user's device, or the user returns to the form to start a new story.

Scenario 5: Using the Demo Studio Without Keys

User Input: A user runs the application without Gemini or image-service keys.

System Processing: The application shows a "Demo studio" notice and generates preview text and sample art locally.

Output: A complete five-frame demo comic, with frames labelled "Sample art" and the story marked as a local demo story.

Technical Architecture & Component Flow

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1. ComicCraft Layered System Architecture

ComicCraft is structured into decoupled tiers so that the interface, the request handling, and the AI services can each change without affecting the others.

2. Component Responsibilities & Relationships

Subsystem Tier	Core Components	Operational Function
Presentation Tier	Jinja2 templates, CSS, JavaScript	Renders the landing page, the story form, and the generated comic view.
API & Routing Tier	FastAPI application on Uvicorn	Serves the pages, validates form input, and dispatches generation requests.
Cloud Intelligence	Google Gemini models	Generates the title, plot, character voice, frame narration, dialogue, and image prompts.
Art Generation	Image-generation service or local demo art	Produces scene art from the image prompts, or sample art in demo mode.
Data / Storage	None (no database in the MVP)	Stories are generated per request and stay on the user's device.

3. End-to-End Response Flow

User fills the form -> browser sends the request to FastAPI -> input validation -> Gemini generates the five-frame story -> art is generated (live service or local demo) -> FastAPI renders the comic view with Jinja2 -> user reads, downloads, or makes another.

System Prerequisites & Environment Configuration

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1. Software & Hardware Prerequisites

Component	Minimum Specification	Recommended Specification
Operating System	Windows 10/11, Ubuntu 20.04+, or macOS 12+	Windows 11 or Ubuntu 22.04 LTS
Python Runtime	Python 3.x with the venv module	A recent stable Python 3 release, 64-bit
Processor (CPU)	Dual-core processor	Quad-core processor
System Memory (RAM)	4 GB	8 GB
Network Connection	Required for live Gemini generation	Broadband connection

2. Python Dependencies (requirements.txt)

The project installs its dependencies from **requirements.txt**. The main libraries are FastAPI, Uvicorn, Jinja2, the Google GenAI SDK, python-dotenv, and HTTPX. Exact versions are pinned in the file so every team member and evaluator installs the same set.

3. Environment Configuration & Secret Management

API keys are kept in a local **.env** file that is read at startup and is never committed to Git:

```
# .env configuration template (do not commit to version control)
GEMINI_API_KEY=your_gemini_api_key_here
# Optional: key for the image-generation service (live art)
```

Live AI keys are optional

The application still starts without these keys. In that case it runs in demo mode with locally generated preview text and sample art.

4. Verification Checklist

- Python virtual environment (venv) created and activated.
- All packages from requirements.txt installed without conflicts.
- GEMINI_API_KEY set in .env for live generation (optional for demo mode).
- Server started and the home page opens at <http://127.0.0.1:8000>.

Project Workflow & Request Lifecycle

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End-to-End Processing Flow

A ComicCraft request follows a six-stage lifecycle, from user input to a finished comic.

Stage	Workflow Phase	Technical Implementation Details
Stage 1	User Input	The user opens the Creator's Desk form and enters the story spark, main character, setting, story mood, and art direction.
Stage 2	Form Submission	The browser submits the form to the FastAPI generation route. The story spark has a 600-character limit, shown by a live counter.
Stage 3	Validation & Routing	FastAPI checks that the required fields (story spark, main character, setting) are present and within limits, then routes the request.
Stage 4	Story Generation	The Gemini model creates the comic title, a five-frame story, narration, dialogue, and an image prompt for every frame.
Stage 5	Art Generation	With a live key, the image service creates the art. Without one, locally generated sample art is used.
Stage 6	Result Rendering	The comic view shows the title, mood and setting, and five frame cards. The user can download the comic or make another.

Workflow Decision & Routing Logic

- If Gemini and image-service keys are configured: generate the story and art live.
- If keys are missing: run the demo studio with locally generated text and sample art, and show a notice to the user.
- If required input is missing or too long: return a clear validation message and do not call any AI service.
- If an external service fails (network or quota): show a controlled error message instead of crashing the application.

Milestone 1: AI Model Selection & Roles

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1. Milestone Objective

Define which AI services create each part of the comic, and how the application keeps working when live services are not available.

2. Model Responsibilities & Division of Labor

AI Component	Assigned Features	Rationale
Google Gemini models	Plot and title, character voice, frame narration, dialogue, image prompts	Strong creative writing and instruction following, and the ability to return a consistent five-frame structure.
Image-generation service (optional)	Scene art for each frame from the image prompts	Turns the written image prompts into comic art when a live key is provided.
Local demo generator	Preview text and sample art without keys	Lets anyone run and test the application without external accounts or network access.

3. Why Responsibilities Are Separated

- **Separation of Concerns:** Writing and illustration are different tasks, so each is handled by the service best suited to it.
- **Reliability:** The demo generator keeps the application usable when keys or internet access are missing.
- **Cost Control:** Live AI calls happen only when keys are configured, so testing and demonstrations do not use API quota.

4. Generation Settings

Generation settings such as the model name, creativity level, and output length are kept in one configuration location so the whole team can tune them without changing route code.

Activity 1.1: Gemini API Configuration & Security

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1. Activity Objective

Safely configure the Google Gemini client, keep secrets out of the codebase, and handle missing or invalid keys gracefully.

2. Implementation Approach

- Keys are loaded from a `.env` file with `python-dotenv` at application startup.
- A single shared client configuration is used by every generation request.
- If no key is found, the application does not crash. It switches to the local demo studio and tells the user that live AI keys are optional.
- Errors from the Gemini SDK are caught and turned into friendly messages without exposing stack traces.

3. Security & Governance Rules

Security Domain	Enforced Standard in ComicCraft
Zero-Hardcoding	API keys never appear in Python source files, HTML templates, or JavaScript files.
Git Commit Hygiene	<code>.gitignore</code> excludes <code>.env</code> so credentials are never committed to GitHub.
Error Sanitization	SDK exceptions are caught and shown to the user as simple, readable messages.
Report Safety	No live API keys or tokens are written in project reports or submissions.

4. Expected Activity Output

With a valid key, generation requests reach Gemini and return live stories. Without a key, the home page shows "Demo studio is ready - Live AI keys are optional" and generation uses the local demo mode.

Activity 1.2: Story Generation Pipeline

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1. Activity Objective

Build the pipeline that turns the user's form inputs into a structured five-frame comic.

2. Pipeline Inputs

Input	Required	Description
The story spark	Yes	A short idea for the story (up to 600 characters).
Main character	Yes	The name of the character the story follows, for example "Fern".
Setting	Yes	Where the story takes place, for example "Moonlit forest".
Story mood	No	The tone of the story, chosen from a dropdown (for example, Adventurous).
Art direction	No	The visual style of the art, chosen from a dropdown (for example, Storybook).

3. Pipeline Output: The Frame Structure

Every generated comic contains a title, a short subtitle describing the mood, art style, and setting, and five frames. Each frame card in the result view contains:

- a frame number and a frame title (for example, "A clue in the wild");
- scene art, with a "Sample art" label when demo art is used;
- an italic narration line that moves the story forward;
- a short tagline for the frame (for example, "A clue is an invitation.");
- a scene description and a line of character dialogue (for example, Fern: "One more step, and we change everything.");
- a collapsible "See image prompt" section showing the text used to create the art.

4. Pipeline Steps

Step	Action	Result
1	Combine the form inputs into a structured prompt	A single instruction that asks for a title and exactly five frames.
2	Send the prompt to the Gemini model	Story text with narration, dialogue, and image prompts.
3	Check and organize the response	Five complete frames, each with all required fields.
4	Generate or attach art	Live art from the image service, or locally generated sample art.
5	Render the comic view	A finished comic page that the user can download.

Activity 1.3: FastAPI Backend Architecture

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1. Activity Objective

Build a modular FastAPI backend with clear routes, input validation, and consistent error handling.

2. Backend Design

- **Application entrypoint:** declares the routes, serves the Jinja2 templates, and mounts static assets.
- **Generation logic:** separate modules handle story generation, art generation, and the local demo mode.
- **Configuration:** a shared module reads the environment settings and AI client configuration.

3. Routes

Route	Purpose
/ (home page)	Shows the landing page and the Creator's Desk story form.
/generate	Receives the form inputs, generates the comic, and shows the result page.

4. Request Lifecycle & Validation

Pipeline Step	Action Performed	FastAPI Mechanism
1. Request ingestion	Receives the submitted form data.	ASGI request handling
2. Validation	Checks required fields and the 600-character limit on the story spark.	Request validation
3. Dispatch	Calls the story and art generation modules.	Python functions called from the route
4. Rendering	Fills the Jinja2 result template with the comic.	Jinja2 templates
5. Error handling	Catches failures and returns a clear message.	Exception handlers

Design Benefit

Keeping routes, generation logic, and configuration in separate modules makes each part easy to test and replace, for example swapping the image service without touching the story logic.

Activity 1.4: Codebase & Development Setup

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1. Activity Objective

Establish a standard project layout, isolate dependencies in a virtual environment, and connect templates and static files to the application.

2. Project Directory Layout

```
ComicCraft-project/  
  main.py           # FastAPI application, routes, validation  
  templates/        # Jinja2 pages: landing page, story form, comic view  
  static/           # CSS styling and front-end assets  
  tests/            # API and module tests  
  requirements.txt   # Python dependencies  
  .env              # Local secrets (not committed to Git)  
  README.md         # Setup and project documentation
```

The layout above follows the structure listed in the Project Executable Files document. Update file names there and here if the repository changes.

3. Environment Setup Procedure

Step	Action	Terminal Command
Step 1	Create a virtual environment	<code>python -m venv venv</code>
Step 2	Activate it	<code>venv\Scripts\activate</code> (Windows) or <code>source venv/bin/activate</code> (Linux/Mac)
Step 3	Install dependencies	<code>python -m pip install -r requirements.txt</code>
Step 4	Configure the API key	Create <code>.env</code> and set <code>GEMINI_API_KEY</code>
Step 5	Run the development server	<code>uvicorn main:app --reload</code>

4. Templates & Static Assets

The application mounts the **static** folder for styling and loads pages from the **templates** folder, so the landing page, the story form, and the comic view all share one consistent design.

Milestone 2: Five Core Comic Creation Functionalities

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1. Milestone Objective

Implement and document the five core capabilities of ComicCraft, including the role, behavior, and output of each.

2. Detailed Capabilities Breakdown

Feature 1: Story Idea Intake

Role: Collect the story spark, main character, setting, story mood, and art direction on the Creator's Desk form.

Behavior: Marks required fields, shows a live character counter (for example, 84 / 600), and offers "Try a fresh spark" for an example story.

Feature 2: Plot and Character Development

Role: Turn the spark into a title and a five-part story arc around the main character. **Engine:** Google Gemini models. **Behavior:** Keeps the character and setting consistent from the first frame to the last.

Feature 3: Panel Script and Dialogue

Role: Write the narration, tagline, scene description, and character dialogue for each of the five frames. **Engine:** Google Gemini models. **Behavior:** Matches the tone to the selected story mood.

Feature 4: Scene Art and Image Prompts

Role: Provide art for every frame and the image prompt behind it. **Engine:** Image-generation service with live keys, or local sample art in demo mode. **Behavior:** Follows the chosen art direction and lets the user open "See image prompt" for each frame.

Feature 5: Download and Iterate

Role: Let users keep and refine their work. **Behavior:** "Download your comic" saves the comic, "Make another" returns to the form, and the footer note explains that the story stays on the device and no account is needed.

Backend Interface Specification

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1. Interface Contract

ComicCraft serves HTML pages from two routes. The generation route accepts the form fields below and returns the rendered comic view.

Route	Input	Output
/ (home page)	None (browser request)	HTML page with the landing section and the story form.
/generate	Story spark, main character, setting, story mood, art direction	HTML comic view: title, subtitle, frame count, creation date, and five frame cards.

2. Input Field Rules

Field	Required	Rule
The story spark	Yes	Text, up to 600 characters. Empty input is rejected.
Main character	Yes	Non-empty text.
Setting	Yes	Non-empty text.
Story mood	No	One option from the dropdown list.
Art direction	No	One option from the dropdown list.

3. Result Page Contents

- Comic title and a subtitle that names the mood, art style, and setting.
- A frame count and the creation date, with a "local demo story" badge when demo mode is used.
- A notice banner in demo mode explaining that Gemini and image-service keys enable live generation.
- Five frame cards, each with the fields listed in Activity 1.2.
- "Download your comic" and "Make another" actions at the top of the page.

AI Prompt Construction & Response Handling

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1. Prompt Design Strategies

The quality of the comic depends on clear instructions. ComicCraft builds its prompts from the form fields using these strategies:

- **Role and goal:** Tell the model it is a comic story writer creating a short comic.
- **User inputs as data:** Insert the story spark, character, setting, mood, and art direction in clearly marked places so they are treated as story material, not as instructions.
- **Fixed structure:** Ask for a title and exactly five frames, each with the same set of fields.
- **Tone control:** Use the story mood to guide word choice, and the art direction to shape the image prompts.
- **Short dialogue:** Ask for brief, in-character lines that fit inside a comic panel.

2. Response Checking

Because model output can vary, the application checks every response before showing it:

- Confirm that five frames are present and that each has a title, narration, dialogue, and an image prompt.
- Remove any extra formatting the model adds around the structured content.
- If a frame is incomplete or the response cannot be read, show a clear message and let the user try again.

3. Image Prompts

Each frame includes an image prompt that describes the scene, the character, the setting, and the art style. The prompt is shown in the "See image prompt" section, which also helps users reuse the prompt in other art tools.

Milestone 3: Web Interface Architecture & Layout

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1. Milestone Objective

Design a warm, clear web interface that guides a user from a first idea to a finished comic in a single flow.

2. UI Design Philosophy & Palette

Design Element	Specification	Purpose
Warm cream background	Soft off-white page and card surfaces	A calm, paper-like reading area that suits a storybook feel.
Deep navy headings	Dark navy text with a serif display style	Strong, readable titles for the hero text and comic titles.
Coral accent	Orange-coral buttons and highlights	Draws attention to key actions such as "Make my comic" and "Download your comic".
Mono labels	Small monospaced labels (for example, FRAME 01 / 05)	Gives the interface a notebook-and-field-note character.

3. Page Layout

The interface has three main views: a landing section with a call to action, the Creator's Desk form, and the generated comic view.

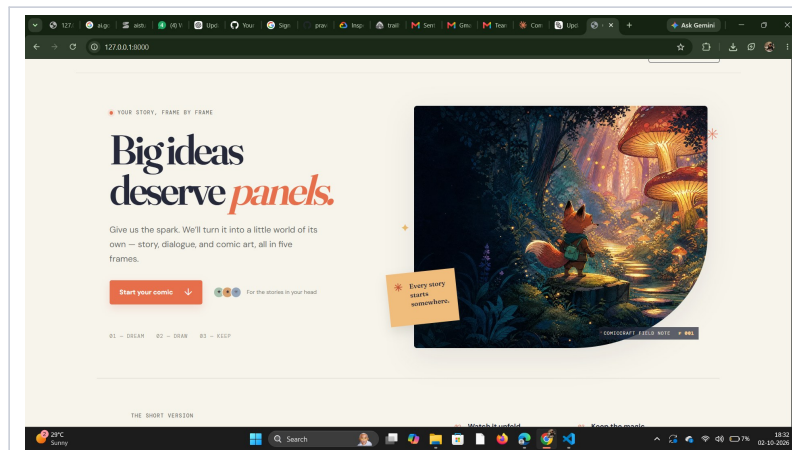


Figure 1 - Landing page: "Big ideas deserve panels." with the Start your comic button and the Dream / Draw / Keep steps.

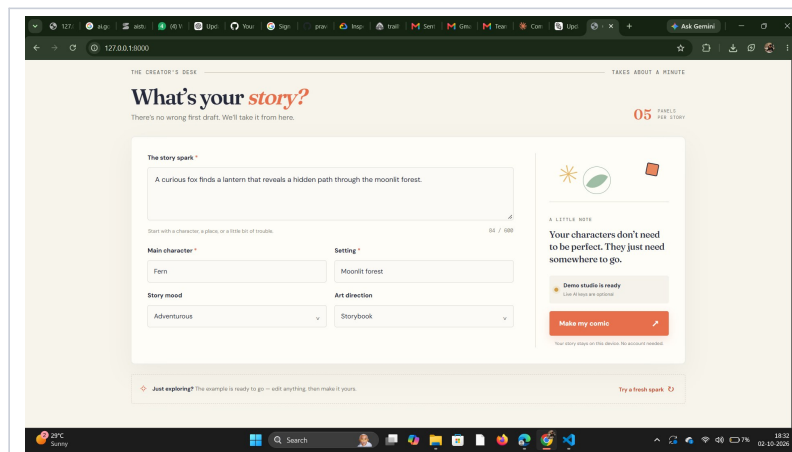


Figure 2 - Creator's Desk: story spark, main character, setting, story mood, art direction, and the Make my comic button.

Dynamic Content Rendering & Interaction

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1. Result Rendering

After the user clicks "Make my comic", the result page shows the comic title, subtitle, and five frame cards. The first frame is shown full width and the following frames are shown in a card grid.

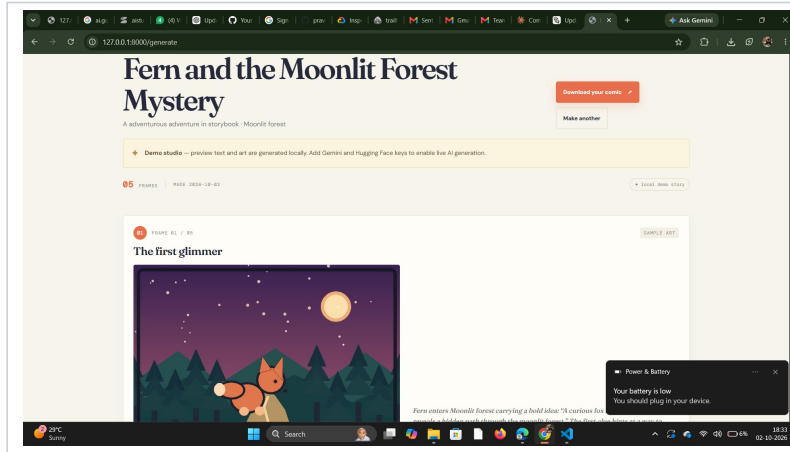


Figure 3 - Generated comic: "Fern and the Moonlit Forest Mystery" with the demo studio notice and the first frame.

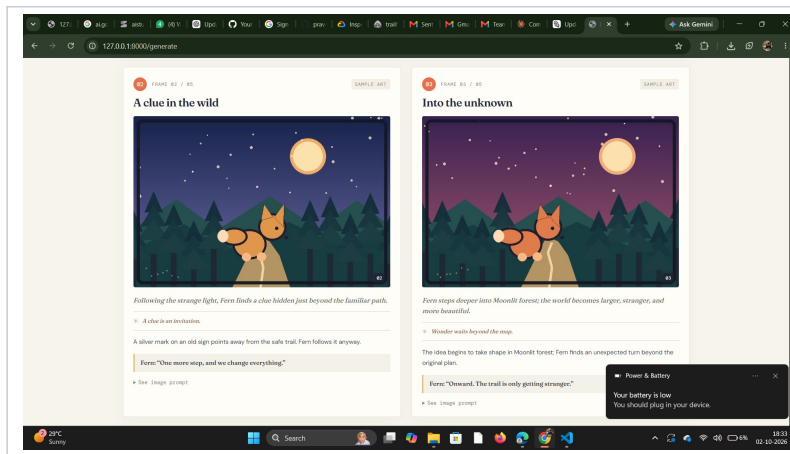


Figure 4 - Frames 2 and 3 with scene art, narration, tagline, dialogue, and the See image prompt section.

2. Interface Interactions

User Action	Interface Response	Purpose
Types in the story spark	The counter updates (for example, 84 / 600).	Keeps the story idea within the limit.
Clicks "Try a fresh spark"	The form is filled with an example story.	Helps users who are not sure where to begin.
Chooses mood or art direction	Dropdown menus show the available options.	Lets users shape the tone and style.
Opens "See image prompt"	A collapsible section shows the prompt text.	Shows how each frame's art was described.
Clicks "Download your comic"	The comic is saved to the device.	Lets users keep their work.
Clicks "Make another"	The user returns to the story form.	Starts a new comic quickly.

Milestone 4: Local Deployment & Operations

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1. Milestone Objective

Run ComicCraft on a local workstation with Uvicorn, verify the environment configuration, and confirm that the application works in demo mode and with live keys.

2. Step-by-Step Deployment Runbook

Step 1: Prepare the environment

```
python -m venv venv
venv\Scripts\activate      # Linux/Mac: source venv/bin/activate
```

Step 2: Install dependencies and configure keys

```
python -m pip install -r requirements.txt
# Create a .env file and set GEMINI_API_KEY (optional for demo mode)
```

Step 3: Launch the server

```
uvicorn main:app --reload
```

Open **http://127.0.0.1:8000** in a browser to see the landing page.

3. Operational Verification

Check	Action	Expected Result
Home page	Open http://127.0.0.1:8000	The landing page and the Creator's Desk form are shown.
Demo mode	Run without API keys and click Make my comic	A five-frame demo comic is shown with a demo studio notice.
Live mode	Set the API keys and generate a comic	A live story is shown, with art from the image service.

4. Common Pitfalls & Remedies

Issue	Remedy
Live generation does not start	Check that GEMINI_API_KEY is set in .env and that the server was restarted.
Frames are labelled "Sample art"	No image-service key is configured. Add the key to use live art.
Port already in use	Stop the other process or start Uvicorn with a different --port value.

System Testing, Verification & Test Matrix

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1. Quality Assurance Strategy

Testing combines automated API and module tests (in the tests folder) with manual walkthroughs of the interface. The matrix below lists the planned checks and the expected result for each. Actual outcomes are recorded when each test is run.

2. Formal Test Matrix

Test ID	Target	Test Condition & Input	Expected Result	Status
TC-01	/ (home page)	Open the home page in a browser	Landing page and form load without errors	To be recorded
TC-02	/generate	Valid story spark, character, and setting	A five-frame comic is shown	To be recorded
TC-03	/generate	Empty story spark	Clear validation message; no comic is generated	To be recorded
TC-04	/generate	Story spark longer than 600 characters	Input is limited or rejected with a message	To be recorded
TC-05	/generate	Missing main character or setting	Clear validation message	To be recorded
TC-06	/generate	Different story mood and art direction	Comic matches the chosen options	To be recorded
TC-07	Demo mode	Run with no API keys	Demo comic with sample art and a notice banner	To be recorded
TC-08	Live mode	Run with valid API keys	Live story and art are returned	To be recorded
TC-09	Actions	Click Download your comic, then Make another	Comic is saved; form is shown again	To be recorded

3. Performance Testing

Response time, throughput, error rate, and resource usage are covered in the Performance Testing document. Those measurements are pending execution, so no numeric results are claimed here.

Security Governance & System Reliability

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1. API Key Protection & Environment Governance

- **No keys in source code:** Keys never appear in .py, .html, or .js files.
- **Environment-only loading:** Keys are read from environment settings at runtime.
- **Safe logging:** Logs and error messages never print API keys.

2. Input Validation & Misuse Defense

Threat	Potential Issue	ComicCraft Countermeasure
Prompt injection	A story spark that tries to override the model's instructions.	User text is placed in marked input sections and treated as story material, not as instructions.
Oversized input	Very long text submitted to the form.	The 600-character limit on the story spark and server-side validation.
Cross-site scripting	Script tags typed into form fields.	Jinja2 escapes dynamic text when rendering pages.

3. Privacy

ComicCraft needs no account. The interface states that the story stays on the user's device, and the current MVP has no database that stores user stories.

4. Error Handling & Graceful Degradation

If the internet connection is lost, a quota limit is reached, or a service returns an unusable response, the application shows a clear message and stays online. Without keys, it falls back to the local demo studio so the interface remains usable.

Maintainability Standard

Following clear module separation and readable code conventions keeps the codebase easy to maintain and extend for future teams.

Project Usage Walkthrough & User Guide

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1. User Walkthrough (Step-by-Step)

Step	Feature	User Actions & System Response
Step 1	Launch	Open http://127.0.0.1:8000 . The landing page loads with the headline "Big ideas deserve panels."
Step 2	Start	Click "Start your comic" to go to the Creator's Desk form.
Step 3	Describe the story	Type a story spark, then enter the main character and setting. Use "Try a fresh spark" if you need an example.
Step 4	Choose style	Pick a story mood (for example, Adventurous) and an art direction (for example, Storybook).
Step 5	Make the comic	Click "Make my comic". The result page shows the title and five frames with art, narration, and dialogue.
Step 6	Keep or repeat	Click "Download your comic" to save it, or "Make another" to start a new story.

2. Submission Checklist

- Complete source code with templates, static files, and tests.
- requirements.txt and README.md with setup instructions.
- Environment configuration (.env template) with no secret keys committed.
- Project documents: Solution Architecture, Problem-Solution Fit, Proposed Solution, Solution Requirements, Technology Stack, Project Planning, Data Flow Diagram, Brainstorming & Idea Prioritization, Performance Testing, and Project Executable Files.
- Demo video or walkthrough (to be provided).

Conclusion & Future Enhancements

ComicCraft - AI Comic Story Creator using Gemini Models • Team ID: SWTID-2026-6777

1. Project Outcomes & Value Delivered

ComicCraft shows how Generative AI can turn a short idea into a complete visual story. By combining Google Gemini models for writing with an art-generation step and a friendly web interface, it gives anyone a guided path from "a spark" to "a finished five-frame comic", with no drawing skills or account required.

2. Summary of Technical Milestones

Milestone Area	Delivered Artifacts	Impact
Backend Engineering	FastAPI application with a home route, a generation route, and input validation.	A simple, reliable way to turn form input into a comic.
AI Generation	Gemini-based story generation with image prompts, plus optional live art and a local demo mode.	Complete comics with or without live API keys.
Web Interface	Landing page, Creator's Desk form, and comic view built with Jinja2 templates and CSS.	A warm, clear experience from idea to finished comic.
Security & QA	Environment-based secrets, input validation, and a planned test matrix.	Safe handling of keys and user input.

3. Strategic Future Enhancements

- **Character Consistency:** Keep a character's look identical across every frame.
- **More Art Styles and Moods:** Expand the dropdown options with more visual styles.
- **Longer Stories:** Allow more than five frames, with chapters.
- **Print-Ready Export:** Export comics as a PDF or an image sheet.
- **Saved Library:** Optional accounts and storage so users can revisit earlier comics.
- **Multilingual Comics:** Generate narration and dialogue in additional languages.

*** End of Sample Project Documentation • ComicCraft - AI Comic Story Creator using Gemini Models • Team ID: SWTID-2026-6777 • 30 September 2026 ***